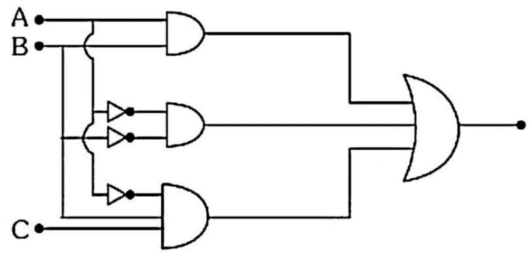
In an experiment, the percentage of error occurred in the measurement of physical quantities A, B, C and D are 1%, 2%, 3% and 4%, respectively. Then the maximum percentage of error in the measurement X, where $X = \frac{A^2 B^{\frac{1}{2}}}{C^{\frac{1}{3}} D^3}$ will be

- $\left(\frac{3}{13}\right)\%$
- 16%
- -10%
- 10%

According to the assumptions made in the kinetic theory of gases, when two molecules of a gas collide with each other then....

- ○ neither kinetic energy nor momentum is conserved.
- ○ momentum is conserved but kinetic energy is not conserved.
- ○ both kinetic energy and momentum are conserved.
- ○ kinetic energy is conserved but momentum is not conserved.

For the given logic circuit; pick the correct option



- For $A = 1,\ B = 0, C = 0$; Output $= 1$
- For $A = 0,\ B = 1, C = 1$; Output $= 1$
- For $A = 1,\ B = 1, C = 0$; Output $= 0$
- For $A = 0,\ B = 1, C = 1$; Output $= 0$

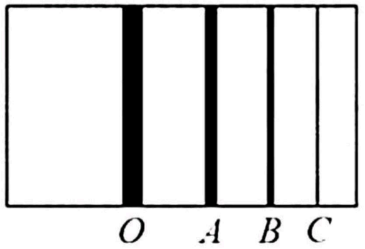
The energy of a photon is equal to the kinetic energy of a proton. If λ_1 is the de-Broglie wavelength of a proton, λ_2 the wavelength associated with the photon and if the energy of the photon is E, then (λ_1/λ_2) is proportional to--

- ○ E^4
- ○ $E^{1/2}$
- ○ E^2
- ○ E

In an electromagnetic wave in free space the root mean square value of the electric field is $E_{rms} = 6\ V/m$.
The peak value of the magnetic field is :-

- ○ $2.83 \times 10^{-8}\ T$
- ○ $0.70 \times 10^{-8}\ T$
- ○ $4.23 \times 10^{-8}\ T$
- ○ $1.41 \times 10^{-8}\ T$

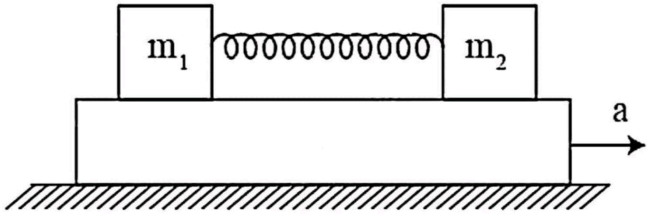
The position of the direct image obtained at O , when a monochromatic beam of light is passed through a plane transmission grating at normal incidence is shown in figure. The diffracted images A, B and C correspond to the first, second and third order diffraction. When the source is replaced by another source of shorter wavelength



:

- all the four will shift in the direction C to O
- all the four will shift in the direction O to C
- the images C,B and A will shift towards O
- the images C,B and A will shift away from O

Two block of masses m_1 and m_2 are connected with a massless unstretched spring and placed over a plank moving with an acceleration ' a ' as shown in figure. The coefficient of friction between the blocks and platform is μ.



- spring will be stretched if $a > \mu g$
- spring will be compressed if $a \leq \mu g$
- spring will neither be compressed nor be stretched for $a \leq \mu g$
- spring will be in its natural length under all conditions

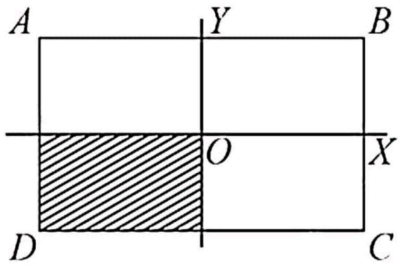
A 2.0 cm object is placed 15 cm in front of a concave mirror of focal length 10 cm. What is the size and nature of the image?

- ○ 4 cm. real
- ○ 4 cm, virtual
- ○ 1.0 cm, real
- ○ None

A particle of mass $1\ \text{mg}$ and charge q is lying at the mid-point of two stationary particles kept at a distance $2\ \text{m}$ when each is carrying same charge q. If the free charged particle is displaced from its equilibrium position through distance x $(x << 1\ \text{m})$. The particle executes SHM. Its angular frequency of oscillation will be _____ $\times 10^8\ \text{rad s}^{-1}$ (if $q^2 = 10C^2$)

- ○ 4
- ○ 3
- ○ 6
- ○ 8

Figure shows a rectangular copper plate with is centre of mass at the origin O and side $AB = 2BC = 2$ m. If a quarter part of the plate (shown as shaded) is removed, the centre of mass of the remaining plate would lie at



- ○ $\frac{1}{12}\,m, \frac{1}{6}\,m$
- ○ $\frac{1}{6}\,m, \frac{1}{12}\,m$
- ○ $\frac{1}{3}\,m, \frac{1}{6}\,m$
- ○ $\frac{1}{3}\,m, \frac{1}{2}\,m$

The engine of a motorboat is switched off when its speed is v_0. If motorboat experiences variable acceleration as $\frac{dv}{dt} = -\beta v^3$, where β is a positive constant and v instantaneous speed, then the speed of motorboat after t second ($t = 0$ when switch is off) is....

- ○ $\frac{v_0}{\sqrt{1+2\beta v_0^2 t}}$
- ○ $\frac{v_0}{\sqrt{1+\beta v_0 t}}$
- ○ $\frac{v_0}{\sqrt{1-\beta v_0^2 t}}$
- ○ $\frac{v_0}{\sqrt{1-\beta v_0 t}}$

In a photodiode, the conductivity increases when the material is exposed to light. It is found that conductivity changes only if the wavelength is less than 600 nm. What is the bandgap?

- ○ 2.06 eV
- ○ 1.2 eV
- ○ 3.4 eV
- ○ 4.2 eV

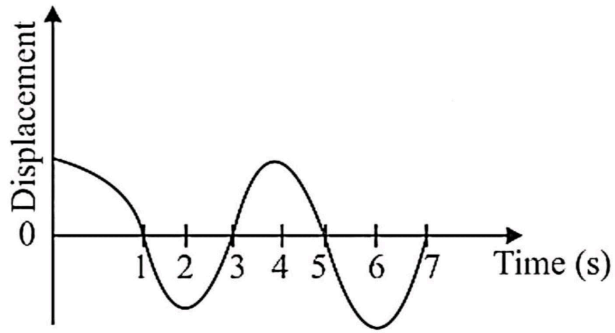
If the magnetic dipole moment of an atom of diamagnetic material, paramagnetic material and ferromagnetic material is denoted by μ_d, μ_p and μ_f respectively, then.

- ○ $\mu_d = 0$ and $\mu_p \neq 0$
- ○ $\mu_d \neq 0$ and $\mu_p = 0$
- ○ $\mu_p = 0$ and $\mu_f \neq 0$
- ○ $\mu_d \neq 0$ and $\mu_f \neq 0$

Which of the following will pair up to produce stationary wave?
(1) $Z_1 = A\cos(kx - \omega t)$
(2) $Z_2 = A\cos(kx + \omega t)$
(3) $Z_3 = A\cos(ky - \omega t)$
(3) $Z_4 = A\cos(kz + \omega t)$

- ○ 1 and 2
- ○ 2 and 3
- ○ 3 and 4
- ○ 1 and 3

A particle is undergoing SHM. Displacement vs time curve is as shown in the figure.



The correct statement is

- Phase of the oscillator is same at $t = 0$ s and $t = 2$ s.
- Phase of the oscillator is same at $t = 2$ s and $t = 5$ s.
- Phase of the oscillator is same at $t = 1$ s and $t = 7$ s.
- Phase of the oscillator is same at $t = 1$ s and $t = 5$ s.

Match **List - I** (Electromagnetic wave type) with **List - II** (Its association/application) and select the correct option from the choices given below the lists :

	List - I		List - II
(a)	Infrared waves	(i)	To treat muscular strain
(b)	Radio waves	(ii)	For broadcasting
(c)	X - rays	(iii)	To detect fracture of bones
(d)	Ultraviolet rays	(iv)	Absorbed by the ozone layer of the atmosphere

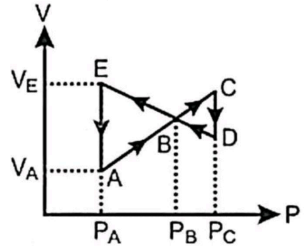
○ **(a) (b) (c) (d)**
(iv) (iii) (ii) (i)

○ **(a) (b) (c) (d)**
(i) (ii) (iv) (iii)

○ **(a) (b) (c) (d)**
(iii) (ii) (i) (iv)

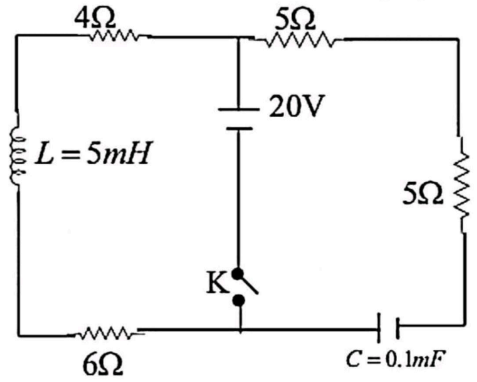
○ **(a) (b) (c) (d)**
(i) (ii) (iii) (iv)

Figure shows a cyclic process ABCDBEA performed on an ideal cycle. If $P_A = 2$ atm, $P_B = 5$ atm and $P_C = 6$ atm, $V_E - V_A = 20$ litre, find the work done (in KJ) by the gas in the complete process. (1 atm. pressure = 1×10^5 Pa)



- ○ 2.67
- ○ 3.18
- ○ 4.27
- ○ 1.37

In the circuit shown, the key (K) is closed at $t = 0$, the current through the key at the $t = 10^{-3} \ln 2$ s; is.



- ○ 2A
- ○ 3.5A
- ○ 2.5A
- ○ Zero

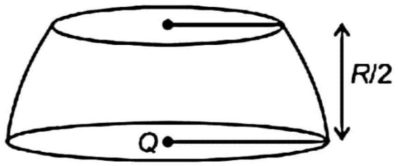
If first excitation potential of a hydrogen like atom is V electron volt, then the ionization energy of this atom will be:

- ○ V electron volt
- ○ $\frac{3\,V}{4}$ electron volt
- ○ $\frac{4\,V}{3}$ electron volt
- ○ cannot be calculated by given information,

Let ω is the angular velocity of the earth's rotation about its own axis and acceleration due to gravity on the earth's surface at the poles is g. An object weighed by a spring balance gives the same reading at the equator, as well as at a height h above the poles $(h \ll R)$. What is the value of h ?

- $\frac{gR}{\omega}$
- $\frac{\omega^2 R^2}{g}$
- $\frac{\omega^2 R^2}{2g}$
- $\frac{2\omega^2 R^2}{g}$

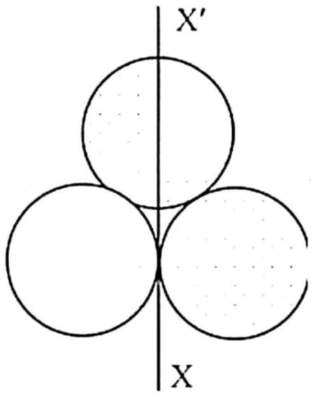
A charge particle having charge Q is placed at the centre of hollow hemispherical sphere whose a portion is removed as shown. If the radius of the hollow sphere is R, then flux through the remaining surface is equal to $Q/x\varepsilon_0$. The value of x is?



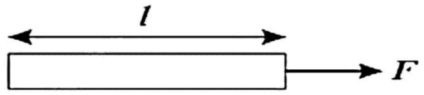
A tuning fork and uniform string of length 100 cm give 4 beats /s. The string is made shorter without any change in tension and the mode of oscillation, until its frequency becomes equal to that of the fork. If now the length of the string is 99 cm, and the frequency of the fork is 100nHz, find the value of n.

An ionization chamber with parallel conducting plates as anode and cathode has 5×10^7 electrons and the same number of singly-charged positive ions per cm^3. The electrons are moving at $0.4\ \mathrm{m\ s}^{-1}$. The current density from anode to cathode is $4\ \mu\mathrm{A\ m}^{-2}$. The velocity of positive ions moving towards cathode is V, find 100V in m/s.

Three identical hollow sphere, each of mass 2kg and radius 10cm are placed as shown in diagram.Consider an axis XX' which is touching to two shells and passing through diameter of third shell. Calculate Moment of inertia of the system consisting of these three spherical shells XX' axis in $g\,m^2$?



A uniform rod of length l is acted upon by a force F in a gravity-free region, as shown in the figure. If the area of cross-section of the rod is A and it's Young's modulus is Y, then the elastic potential energy stored in the rod due to elongation is $U = \frac{F^2 l}{\mathrm{n}AY}$. Then find the value of n?



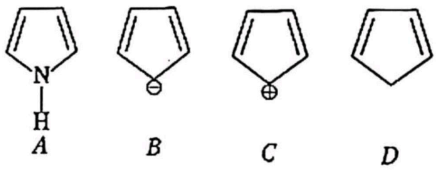
The following pairs of compounds can be distinguished by which of the following tests:

(i) Ethylamine and aniline

(ii) Aniline and N-methylaniline

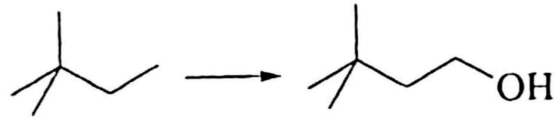
- ○ i) Hinsberg's reagent, and ii) Azo dye test
- ○ i) with Hinsberg's reagent, and ii) carbyl amine test
- ○ i) Carbyl amine test, and ii) Azo dye test
- ○ i) Azo dye test, and ii) Carbyl amine test

Which of the following compounds are aromatic?



- ○ *A*, *B*
- ○ *A*, *B*, *C*
- ○ *B*, *C*
- ○ *B*, *C*, *D*

What sequence of reagents is required to accomplish the following transformation ?



(i) NBS, ROOR ; (ii) $CH_3CH_2\overset{\ominus}{O}$

○ (iii) $N\overset{\ominus}{H}_2$;

(iv) Disiamyl borane (v) H_2O_2, $O\overset{\ominus}{H}$

○ (i) Cl_2/hv (ii) $O\overset{\ominus}{H}$, heat, (iii) 2HCl

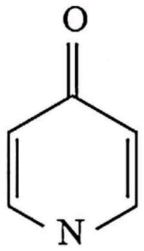
(iv) $O\overset{\ominus}{H}$, heat ; (v) $HgSO_4$, H_2SO_4

○ (i) NBS, ROOR ; $O\overset{\ominus}{H}$, DMSO

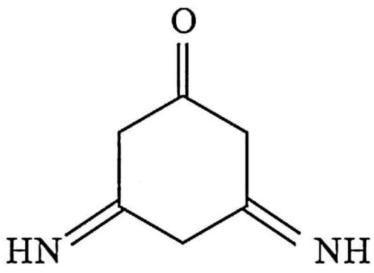
(i) Br_2, hv ; (ii) t-butoxide ; (iii)

○ BH_3. THF, H_2O_2, $O\overset{\ominus}{H}$

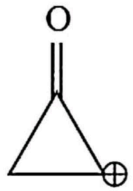
how many of the following enolic form is more stable than keto form



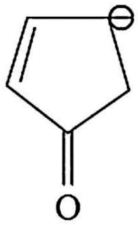
(I)



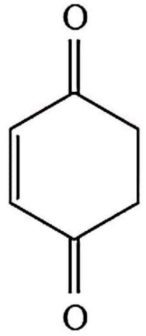
(II)



(III)



(IV)

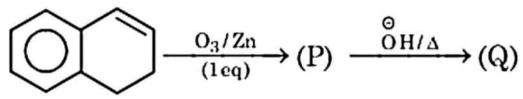


(V)

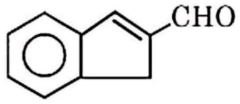
- ○ 2
- ○ 3
- ○ 4
- ○ 1

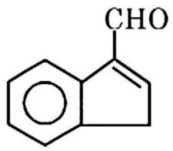
Arrange the following in increasing order of melting point:
Li_2O, LiF, Li_3N

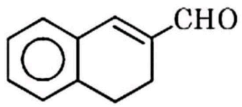
- ○ $Li_3N < Li_2O < LiF$
- ○ $Li_2O < Li_3N < LiF$
- ○ $LiF < Li_2O < Li_3N$
- ○ $Li_3N < LiF < Li_2O$

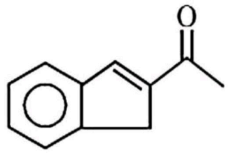


is :

○ 

○ 

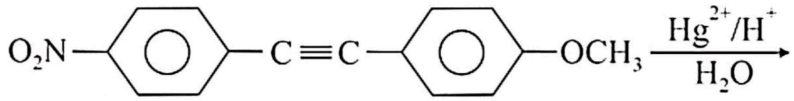
○ 

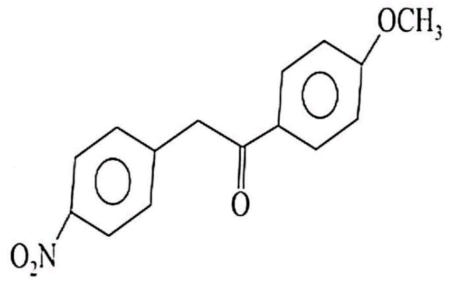
○ 

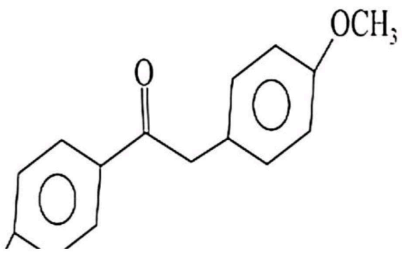
Identify the incorrect order corresponding the mentioned property.

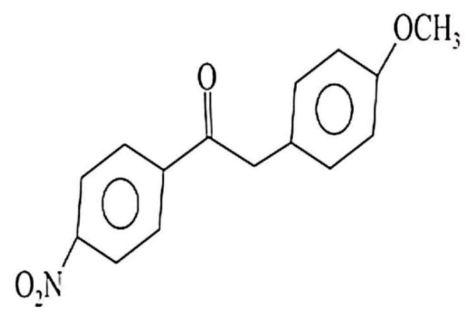
- $Al_2O_3 < MgO < Na_2O < K_2O$: Basicity
- $NH_3 < PH_3 < AsH_3$: Acidity
- $Li^+ < Na^+ < K^+ < Cs^+$: Ionic radius
- $Li < Be < B < C$: First ionisation enthalpy

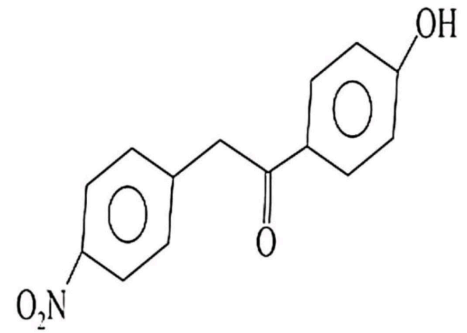
The major product obtained from the following reaction is :



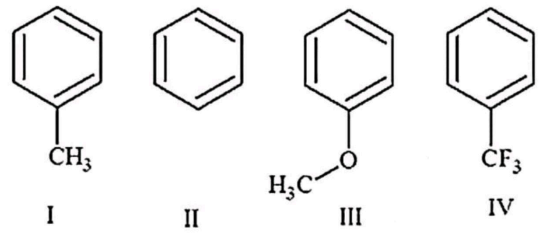
○ 

○ 

○ 
OCH3
O
O2N

○ 
OH
O
O2N

Among the following compounds, the decreasing order of reactivity towards electrophilic substitution is



I II III IV

- ○ III > I > II > IV
- ○ IV > I > II > III
- ○ II > I > IV > III
- ○ I > III > II > IV

A purple colored compound of Manganese (X) decomposes on heating to liberate oxygen and forms compounds of Manganese (Y and Z) compound Z reacts with KOH in presence of potassium nitrate to give compound Y. Compounds X, Y and Z respectively are-

- $X - KMnO_4$ $Y - Mn_2O_7$ $Z - MnO_2$
- $X - K_2MnO_4$ $Y - KMnO_4$ $Z - Mn_2O_7$
- $X - KMnO_4$ $Y - K_2MnO_4$ $Z - MnO_2$
- $X - KMnO_4$ $Y - MnO_2$ $Z - MnO$

n, l and m values of orbitals A and B are given below.

	n	l	m
A	3	2	0
B	3	1	0

Which statement about A and B orbitals are correct?

(a) A has more energy than B.

(b) A possess electron density along axial direction only.

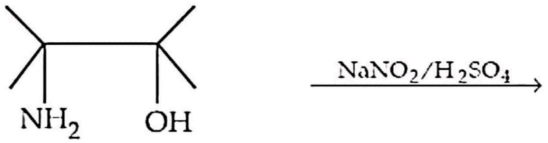
(c) B has one angular node and two radial nodes.

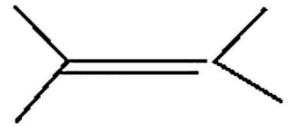
(d) Probability of finding electron density in B is zero along $x - z$ plane.

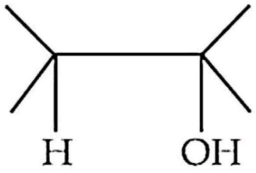
Assume that for d sub-shell, $m = 0$ for d_{xz} orbital and for p sub-shell, $m = 0$ for p_y orbital.

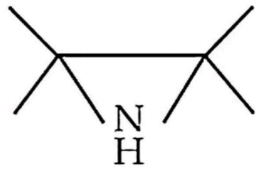
- ○ a and b
- ○ b and c
- ○ c and d
- ○ a and d

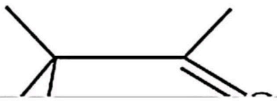
The major product of the reaction is



○ 

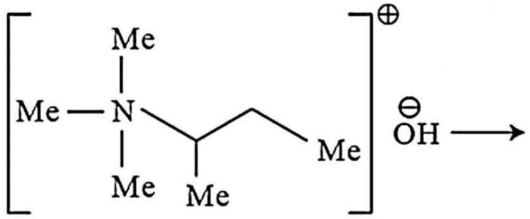
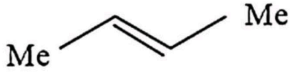
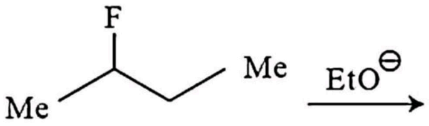
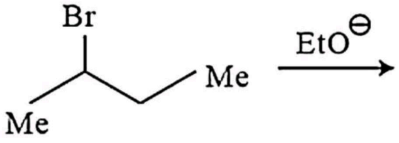
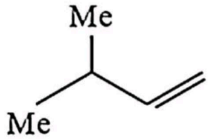
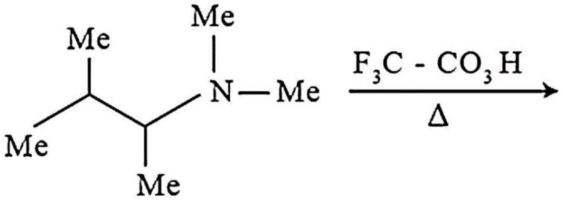
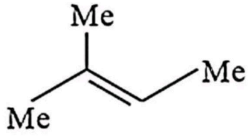
○ 

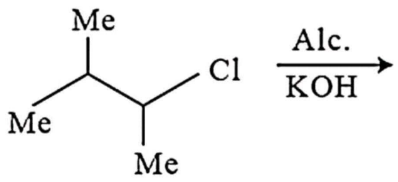
○ 

○ 

A compound absorbs light in the wavelength region $490 - 500$ nm. Its complementary colour is-

- ○ red
- ○ blue
- ○ orange
- ○ blue-green

	Column I		Column II
S. No.	Compound		Product
a.			p. 
b.			q. 
c.			r. 
d.			s. 

e. 

- $a \rightarrow p; b \rightarrow q; c \rightarrow s; d \rightarrow r; e \rightarrow s$
- $a \rightarrow q; b \rightarrow q; c \rightarrow p; d \rightarrow r; e \rightarrow s$
- $a \rightarrow q; b \rightarrow p; c \rightarrow p; d \rightarrow r; e \rightarrow s$
- $a \rightarrow r; b \rightarrow q; c \rightarrow s; d \rightarrow p; e \rightarrow q$

Calcium lactate is a salt of weak acid i.e., lactic acid having general formula $Ca(LaC)_2$. Aqueous solution of salt has $0.3M$ concentration. pOH of solution is 5.60. If 90% of the salt is dissociated then what will be the value of pK_a ?

- $2.8 - \log(0.54)$
- $2.8 + \log(0.54)$
- $2.8 + \log(0.27)$
- None of these

The formation of oxide ion O^{2-} from oxygen atom requires an exothermic reaction, followed by an endothermic step as shown below. The process of formation of O^{2-} in gas phase is unfavorable $(\Delta H^{\ominus} = +\text{ ve})$, even though it has stable configuration of the nearest noble gas neon. This is because

$O(g) + e^- \longrightarrow O^-(g); \Delta H^{\ominus} = -141 \text{ kJ mol}^{-1}$

$O^-(g) + e^- \longrightarrow O^{2-}(g); \Delta H^{\ominus} = +760 \text{ kJ mol}^{-1}$

- ○ O^{2-} has larger size than that of neon
- ○ oxygen is more electronegative than neon
- ○ O^{2-} has larger size than oxygen atom
- ○ electron repulsion in oxide ion is more which over comes the stability achieved by noble gas configuration.

Find incorrect statement for coordination compounds.

- $[PtCl_2(NH_3)_2]$ & $[Cu(NH_3)_4]^{2+}$ are square planar complexes.
- Complex $[Fe(C_2O_4)_3]^{3-}$, does not exhibit geometrical isomerism while exhibits optical isomerism.
- CO bond order : $[Ni(CO)_4] > [Co(CO]_4]^- > [Fe(CO)_4]^{2-}$.
- $[Ni(Cl_2)(PPh_3)_2]$ has two geometrical isomers.

Predict the correct order of the metallic character among the following:

- ○ Cu > Ag > Au > Rg
- ○ Rg > Au > Ag > Cu
- ○ Au > Ag > Cu > Rg
- ○ Rg > Cu > Au > Ag

Assertion : A mixture of plant pigments can be separated by chromatography.
Reason: Chromatography is used for the separation of coloured substances into individual components.

- ○ If both assertion and reason are true and the reason is the correct explanation of the assertion.
- ○ If both assertion and reason are true but reason is not the correct explanation of the assertion.
- ○ If assertion is true but reason is false.
- ○ If the assertion and reason both are false.

The time taken by a first order reaction for the completion upto 99% was 32 min. Then calculate time taken by same reaction for completion upto 99. 9%.

- ○ 24 min
- ○ 8 min
- ○ 4 min
- ○ 48 min

Standrad entropies of X_2, Y_2 and XY_3 are 60, 40 and 50 $JK^{-1}mol^{-1}$ respectively. For the reaction $\frac{1}{2}\,X_2 + \frac{3}{2}\,Y_2 \leftrightarrow XY_3, \Delta H = -30\ kJ$ to be at equilibrium, the temperature should be ____ K

Arginine is nonessential amino acid which is present is human body. How many Nitrogen atoms are present in sp^2 hybrid state in Arginine.

K_{sp} of SrF_2 is 1×10^{-10} . The solubility of SrF_2 in 0.1 M NaF is 1×10^{-a} M. The value of a is.....

In the crystalline solid $MSO_4 \cdot nH_2O$ of molar mass 250 g mol^{-1} , the percentage of anhydrous salt is 64 by weight. The value of n is-

A weak electrolyte having the limiting equivalent conductance of $400\ \mathrm{S\ cm^2\ g\ equivalent^{-1}}$ at $298\ \mathrm{K}$ is 2% ionized in its $0.1\ \mathrm{N}$ solution. The resistance of this solution (in ohms) in an electrolytic cell of cell constant $0.4\ \mathrm{cm^{-1}}$ at this temperature is....

If P is an orthogonal matrix and $Q = PAP^T$ and $B = P^TQ^{1000}P$, then B^{-1} is, where A is involutory matrix____

- A
- A^{999}
- I
- None of these

Let S be the set of real values of λ for which the function $f(x) = x^3 - 3(2\lambda - 1)x^2 + 6\lambda x$ has exactly one local maximum and exactly one local minimum, then S can be

- ○ $(0, 6)$
- ○ $(1, 4)$
- ○ $(-\infty, 0)$
- ○ $(0, \infty)$

If the system of equation

$$\begin{pmatrix} 1 & -2 & 5 \\ 2 & -1 & 1 \\ 11 & -7 & p \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ q \end{pmatrix}$$

has infinitely many solutions, then -

- ○ $p+q=2$
- ○ $p+q=10$
- ○ $p-q=2$
- ○ none of these

Let $f : R \to R$ be defined as

$$f(x) = \begin{cases} -55x, & \text{if } x < -5 \\ 2x^3 - 3x^2 - 120x, & \text{if } -5 \leq x \leq 4 \\ 2x^3 - 3x^2 - 36x - 336, & \text{if } x > 4 \end{cases}$$

Let $A = \{x \in R : f \text{ is increasing}\}$. Then A is equal to:

- ○ $(-5, \infty)$
- ○ $(-5, -4) \cup (4, \infty)$
- ○ $(-\infty, -5) \cup (-4, \infty)$
- ○ $(-\infty, -5) \cup (4, \infty)$

There are two possible values of A in the solution of the matrix equation $\begin{bmatrix} 2A+1 & -5 \\ -4 & A \end{bmatrix}^{-1} \begin{bmatrix} A-5 & B \\ 2A-2 & C \end{bmatrix} = \begin{bmatrix} 14 & D \\ E & F \end{bmatrix}$ where A, B, C, D, E, F are real numbers. The absolute value of the difference of these two solutions of A is

- ○ $\frac{8}{3}$
- ○ $\frac{11}{3}$
- ○ $\frac{1}{3}$
- ○ $\frac{19}{3}$

Suppose $\begin{vmatrix} f'(x) & f(x) \\ f''(x) & f'(x) \end{vmatrix} = 0$, where $f(x)$ is continuously differentiable function with $f'(x) \neq 0$ and satisfies $f(0) = 1$ and $f'(0) = 2$, then $\lim_{x \to 0} \frac{f(x)-1}{x}$ is

- ○ 1
- ○ 2
- ○ $\frac{1}{2}$
- ○ 0

Let $a = 3^{\frac{1}{223}} + 1$ and for all $n \geq 3$, let $f(n) = {}^{n}C_0 \cdot a^{n-1} - {}^{n}C_1 \cdot a^{n-2} + {}^{n}C_2 \cdot a^{n-3} - \ldots\ldots + (-1)^{n-1} \cdot {}^{n}C_{n-1} \cdot a^0$. I
the value of $f(2007) + f(2008) = 9k$ where $k \in N$, then find k.

- ○ 2562
- ○ 3146
- ○ 2275
- ○ 2187

The sum of 50 observations and sum of their squares were found to be 245 and 1401 respectively, but later on, three observations 2, 3 and 5 were found to be not correct. If the incorrect observations are removed, then variance of remaining observations is

- ○ 4.25
- ○ 4.00
- ○ 3.15
- ○ 3.00

The general solution of the differential equation $\frac{dy}{dx} = 2ytanx + tan^2x,\ \forall x \in \left(0, \frac{\pi}{2}\right)$ is $yf(x) = \frac{x}{2} - \frac{\sin(2x)}{4} + C$, (where, C is an arbitrary constant). If $f\left(\frac{\pi}{4}\right) = \frac{1}{2}$, then the value of $f\left(\frac{\pi}{3}\right)$ is equal to

- ○ $\frac{1}{2}$
- ○ $\frac{1}{4}$
- ○ 2
- ○ 4

The area inside the parabola $5x^2 - y = 0$ but outside the parabola $2x^2 - y + 9 = 0$ is equal to: (sq. unit)

- ○ $6\sqrt{3}$
- ○ $12\sqrt{3}$
- ○ $3\sqrt{3}$
- ○ $24\sqrt{3}$

The value of $\sin^{-1}\cos\left(\cos^{-1}\frac{3}{5}+2\tan^{-1}\frac{1}{3}\right)$ is

- ○ $\frac{\pi}{3}$
- ○ $\frac{\pi}{4}$
- ○ 0
- ○ $\frac{\pi}{2}$

If a directrix of a hyperbola centered at the origin and passing through the point $(4, -2\sqrt{3})$ is $5x = 4\sqrt{5}$ and its eccentricity is e, then:

- ○ $4e^4 + 8e^2 - 35 = 0$
- ○ $4e^4 - 24e^2 + 35 = 0$
- ○ $4e^4 - 24e^2 + 27 = 0$
- ○ $4e^4 - 12e^2 - 27 = 0$

Let $A(5, 12), B(-13\cos\theta, 13\sin\theta)$ and $C(13\sin\theta, -13\cos\theta)$ are angular points of $\triangle ABC$ where $\theta \in R$. The locus of orthocentre of $\triangle ABC$ is

- $x - y + 7 = 0$
- $x - y - 7 = 0$
- $x + y - 7 = 0$
- $x + y + 7 = 0$

If two different numbers are taken from the set $\{0, 1, 2, 3, \ldots, 10\}$, then the probability that their sum as well as absolute difference are both multiple of 3 is:

- ○ $\frac{12}{55}$
- ○ $\frac{14}{45}$
- ○ $\frac{7}{55}$
- ○ $\frac{6}{55}$

$\triangle ABC$ is formed by $A(1, 8, 4), B(0, -11, 4)$ and $C(2, -3, 1)$. If D is the foot of the perpendicular from A to BC. Then the coordinates of D are

- ○ $(-4, 5, 2)$
- ○ $(4, 5, -2)$
- ○ $(4, -5, 2)$
- ○ $(4, -5, -2)$

Let $f(x) = x^2 - x + k - 2, k \in R$, then the complete set of values of k for which $y = |f(|x|)|$ is non-derivable at 5 distinict points is

- $(1, 4)$
- $\left(0, \frac{9}{4}\right)$
- $(-\infty, 2)$
- $\left(2, \frac{9}{4}\right)$

If $f : R/\{0\} \to R$ is such that $2f(x) + f\left(\frac{1}{x}\right) = 4x$ and $S = \{x \in R : f(x) = f(-x)\}$, then the number of elements in S is

- 0
- 1
- 2
- at least three

Let $A \equiv (-4, 0)$ and $B \equiv (4, 0)$ if the number of points C on the circle $x^2 + y^2 = 16$, such that the area of triangle whose vertices are A, B and C is positive integer is N then value of $\left[\frac{N}{8}\right]$ is equal to ? (where [] denotes the greatest integer function)

- ○ 4
- ○ 5
- ○ 3
- ○ 7

The coefficient of x^2 in the expansion of $\left(1-x+2x^2\right)\left(x+\frac{1}{x}\right)^{10}$ is

- ○ 210
- ○ 714
- ○ 504
- ○ 240

$\int_{-1}^{1}\{x^2+x-3\}dx$, where $\{x\}$ denotes the fractional part of x, is equal to

- ○ $\frac{1}{3}(1+3\sqrt{5})$
- ○ $\frac{1}{6}(1+3\sqrt{5})$
- ○ $\frac{1}{3}(3\sqrt{5}-1)$
- ○ $\frac{1}{6}(3\sqrt{5}-1)$

Let $a_1, a_2, a_3, \ldots\ldots\ldots a_n$ be a finite arithmetic sequence with $a_4 + a_7 + a_{10} = 17$ and $a_4 + a_5 + a_6 + \ldots\ldots + a_{14} = 77$.
if $a_k = 13$, then value of $\frac{k}{2}$ is

If the general solution of the differential equation $(y-x+1)dy-(y+x+2)dx=0$ is $f(x,y,c)=0$, then the value of c such that $f(1,1,c)=0$ is

P, Q, R, S are the points $(1, 2, -2)$, $(8, 10, 11)$, $(1, 2, 3)$ and $(3, 5, 7)$ respectively. If s denotes the projection of PQ on RS then $29s^2 = 100k^2$, absolute value of $k =$

A variable line moves in such a way that the product of the perpendiculars from $(4,\ 0)$ and $(0,\ 0)$ is equal to 9. The locus of the foot of the perpendicular from $(0,\ 0)$ on the variable line is a circle, the square of whose radius is

Let $F(z) = \frac{z+i}{z-i}$ for all complex numbers $z \neq i$, and let $z_n = F(z_{n-1})$ for all positive integers n. Given that $z_0 = \frac{1}{137} + i$ and $z_{2002} = a + bi$, where a and b are real numbers, find unit digit of $a + b$.